

```
def is_safe(vertex, graph, path, position):
```

```
    # Check whether there is an edge
    if graph[path[position - 1]][vertex] == 0:
        return False
```

```
    # Check whether vertex is already included
    if vertex in path:
        return False
```

```
    return True
```

```
def hamiltonian_cycle(graph, path, position):
```

```
    n = len(graph)
```

```
    if position == n:
```

```
        if graph[path[position - 1]][path[0]] == 1:
            return True
```

```
        return False
```

```
    for vertex in range(1, n):
```

```
        if is_safe(vertex, graph, path, position):
```

```
            path[position] = vertex
```

```
            if hamiltonian_cycle(graph, path, position + 1):
                return True
```

```
            path[position] = -1
```

```
    return False
```

```
graph = [
    [0, 1, 1, 0],
    [1, 0, 1, 1],
    [1, 1, 0, 1],
    [0, 1, 1, 0]
]
```

```
n = len(graph)
```

```
path = [-1] * n
```

```
path[0] = 0
```

```
if hamiltonian_cycle(graph, path, 1):
```

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56)
on win32
```

```
Enter "help" below or click "Help" above for more information
```

```
>>>
```

```
==== RESTART: C:/Users/ROOPA SAI DIVYA/OneDrive/Documents/B
Hamiltonian Cycle:
```

```
0 -> 1 -> 3 -> 2 -> 0
```

```
>>>
```